

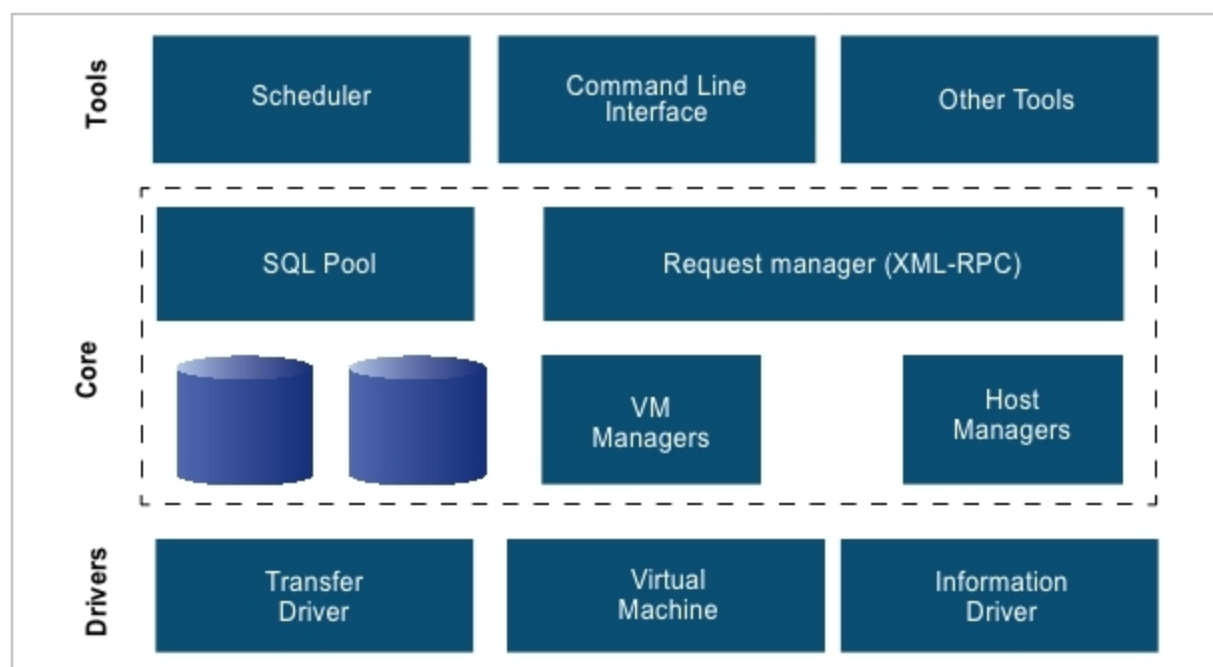


Fig. 3: OpenNebula components

The main features are:

- Offers insights through discovery, monitoring, utilisation, performance, reporting, analytics, charge-back and trending.
- Controls security, compliance, alerting, policy-based resource and configuration management.
- Automates IT processes, tasks and events, provisioning, workload management and orchestration.
- Integrates systems management, tools and processes, event consoles, CMDB, RBA and Web services.

Official website: <http://manageiq.org>

Latest version: Hammer-10

4. Cloudify. Cloudify is an open source software cloud and NFV orchestration product that uses OASIS TOSCA technology. It is designed using Python. Cloudify allows users to model and automate an application's entire life cycle. This includes deployment to a cloud or data centre environment, the management of the deployed application, failure detection and ongoing maintenance. The platform is ideal for users who want to launch prebuilt applications in the cloud without handling the technical aspects.

Cloudify performs the following tasks.

- **Application modelling.** This describes the application with all its resources.
- **Orchestration.** This maintains and runs an application, and performs ongoing operations such as scaling, healing and maintenance.
- **Pluggability.** This provides reusable component abstraction for

the system.

- **Security.** This provides secure communication via SSL, which enables clients to ensure that the data set received is encrypted.

The main features are:

Easy orchestration, built-in node types, a blueprints catalogue, role-based access control, IT security and governance, network and TOSCA orchestration, new NFV capabilities, custom widgets and LDAP integration.

Official website: <http://www.cloudify.co>

Latest version: 5.0

5. Mist.io. Mist.io is a platform that simplifies cloud management and helps users prevent vendor and complexity lock-in. It offers cost and usage reporting, RBAC, management, provisioning, orchestration, monitoring and automation for servers across public and private clouds, Docker containers and KVM hypervisors. It gives actionable alerts so users can address operational issues from anywhere, using any Web-connected device.

The main features are:

- Controls public and private clouds, containers, bare metal servers and more.
- Has fine grained controls for delegating access to team members.
- Enables cost and usage reporting across the whole infrastructure.
- Orchestrates repeatable deployments, and automates common responses.
- Enforces policies consistently, across any computing platform.

Official website: <http://www.mist.io>

6. VirtEngine. VirtEngine is an open source CMP that can be used to build private or public clouds, which support IaaS, PaaS and SaaS. This Platform as a Service system allows customers to deploy applications in a few clicks. VirtEngine has a wide range of applications and a simple user interface for customers to self-serve their needs. It helps users build both public and private clouds within very little time, and supports infrastructure platforms and other storage devices. It is also very scalable and comes with automation tools that provide companies a competitive advantage.

The main features are:

- Access control, demand and supply monitoring, cost management, multi-cloud management, one-click apps, and automatic launch.
- DNS support, self-healing, cloud-native, multi-locations, Docker containers, cloud virtual machines and migration tools.

Official website: <http://www.virtengine.com>

7. openQRM. openQRM is a free and open source cloud computing management platform for managing heterogeneous data centre infrastructures. It provides a complete, automated workflow engine for all bare metal and VM deployment, as well as for all IT sub-systems, enabling professional management and monitoring of the data centre and cloud capacities.

openQRM provides a Web-based, open source data centre management and cloud platform with the help of which various internal and external technologies can be abstracted and grouped within a common management tool. This management system also takes care of provisioning, high availability and the monitoring of services offered. Instead of providing individual tools for individual tasks, such as configuration management and system monitoring, openQRM integrates proven open source management tools such as Nagios and Zabbix.

The main features are:

- Supports P2V, P2P, V2P, V2V migrations and high availability.
- Integrates with all major open and commercial storage technologies.
- Integrated billing system that maps CCU/h (cloud computing units) to real currency.